

Web Programming

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Web programming foundations

Front-end development tools, frameworks & libraries

Developer's tools

Lecture: Developer's tools

Module: Front-end development tools, frameworks & libraries

Developer's tools

- **Developers use a number of tools to get their work done.**
 - ✓ The **code editor** is the **main tool** of a **developer**.
- In recent years, **Visual Studio Code** has become **the most popular tool for coding web projects**.
 - ✓ But there are **tons** of other **options**.
 - ✓ Some popular options include **Sublime Text, Brackets**, etc..

Code editors & IDEs

- **Code editors** are sometimes called **Integrated Development Environments (IDEs)**.
 - ✓ These **terms** can be **used interchangeably** although refer to different things.

Code editors & IDEs

- **IDE**

- ✓ IDEs tend to be more structured, so that things are done in a specific way with lots of shortcuts that help you program faster.
- ✓ IDEs provide hints for developing and structuring your code.
- ✓ An IDE is usually meant for more than web languages.
- ✓ Heavy-weight.

- You can really see this in the difference between an IDE like **Visual Studio** and a basic code editor like **Visual Studio code**.

Desirable features in a code editor or IDE

- Ability to guess the language you're working on and provide assistance in writing the code.
- **Must** have **built-in** references that let you look up documentation.
- **Editors** must check your code for potential errors.
- An **editor** must catch formatting issues, suggest best practices and help your **team** use similar styles.

Visual Studio Code - Code Editing. Redefined

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Visual Studio Code Docs Updates Blog API Extensions FAQ Search Docs Download

Version 1.41 is now available! Read about the new features and fixes from November.

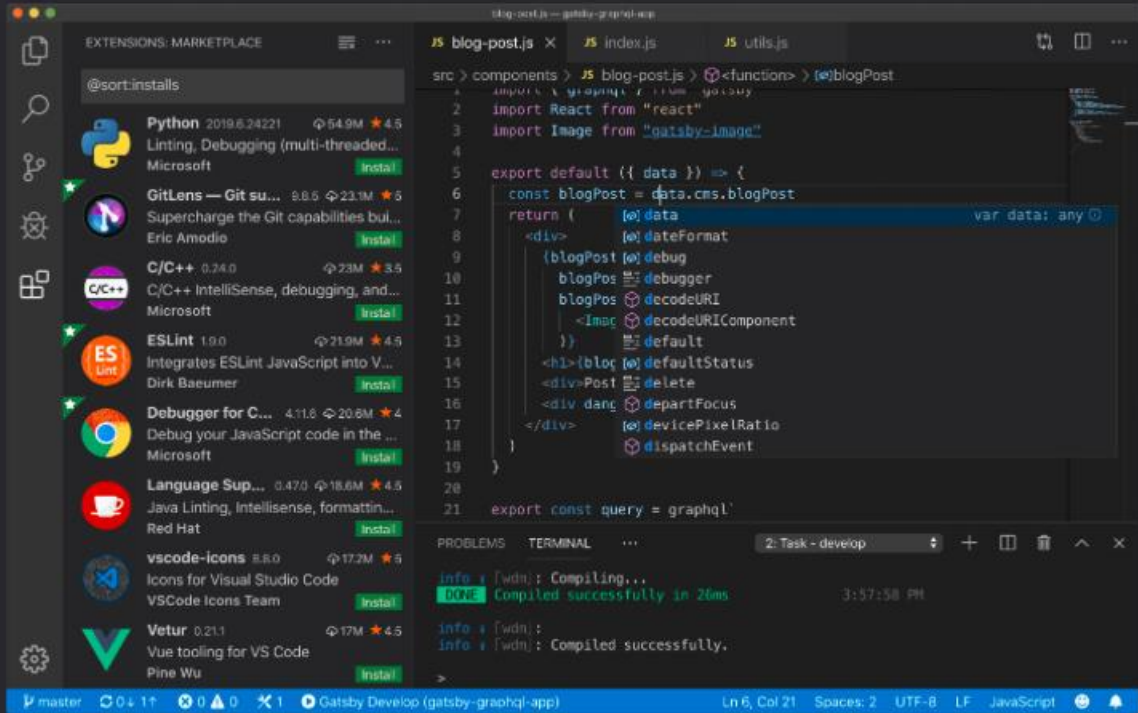
Code editing. Redefined.

Free. Built on open source. Runs everywhere.

[Download for Mac](#)
Stable Build

[Other platforms and Insiders Edition](#)

By using VS Code, you agree to its [license and privacy statement](#).



The screenshot shows the Visual Studio Code interface. On the left is the Extensions Marketplace with a list of extensions including Python, GitLens, C/C++, ESLint, Debugger for C++, Language Support for Java, vscode-icons, and Vetur. The main editor area shows a React component named `blogPost` in `blog-post.js`. The component uses `useState` and `useEffect` hooks. The terminal at the bottom shows the output of a build process, indicating successful compilation.



IntelliSense



Debugging

<https://code.visualstudio.com>

A sophisticated text editor for code, markup and prose

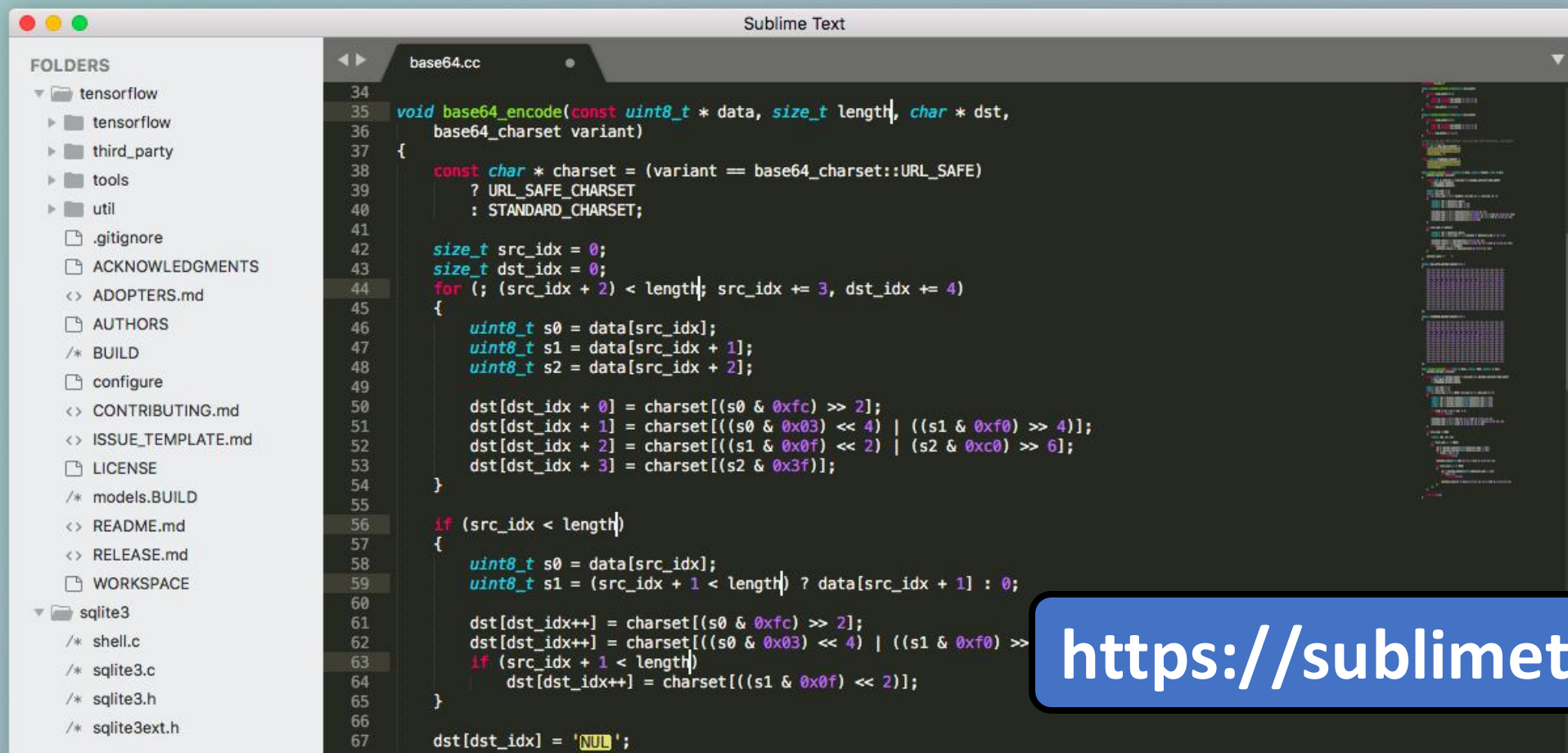
DOWNLOAD FOR MAC

Sublime Text 3 (Build 3211)

[Changelog](#)

Introducing our Git client

[Sublime Merge](#)



<https://sublimetext.com>

Frameworks & libraries

Lecture: Frameworks & libraries

Module: Front-end development tools, frameworks & libraries

Beyond HTML, CSS & JavaScript

- **Developers** approach **how** they **program** in a variety of ways, and some of these are related to the **specific language** they **use**.
- There are **libraries, frameworks** that **complement HTML, CSS and JavaScript**.
- **Why** would I want to **write code** in a **language other** than these **three**?

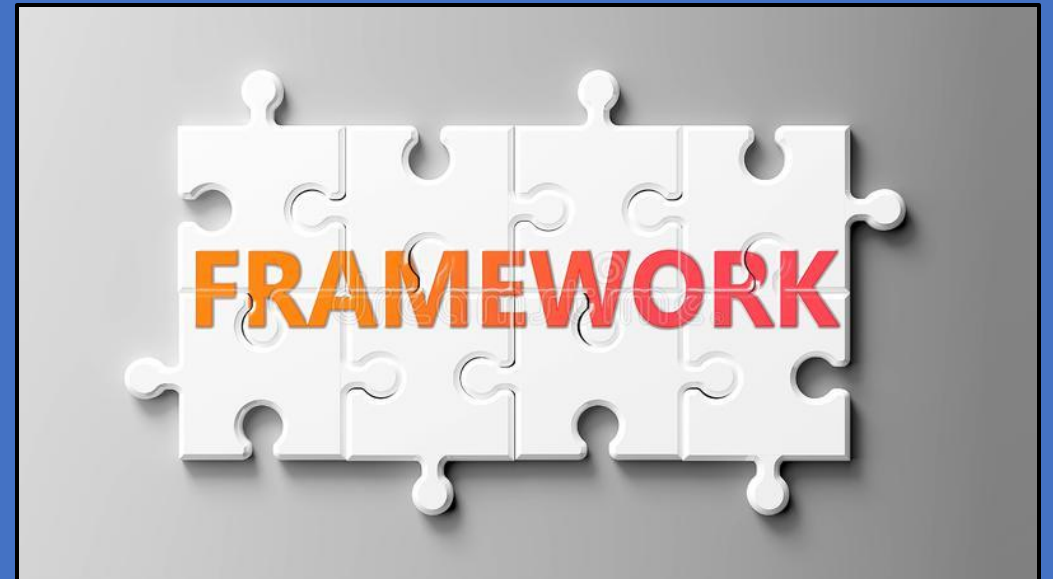
Why would you want to use a library or framework in addition to HTML, CSS, and JavaScript?

The **reason** is simple: **other languages offer more efficient solutions.**

Frameworks & libraries extend the capabilities of existing languages, helping you build projects quicker.

Frameworks & libraries

- Frameworks & libraries usually have been battle tested over time.
- These tools use some of the industry's best practices.
 - ✓ They can also help you learn how professional developers do things.



Source: <https://bit.ly/3QYE32V>



Lodash

A modern JavaScript utility library delivering modularity, performance & extras.

Documentation

FP Guide

```
_.defaults({ 'a': 1 }, { 'a': 3, 'b': 2 });  
// → { 'a': 1, 'b': 2 }  
_.partition([1, 2, 3, 4], n => n % 2);  
// → [[1, 3], [2, 4]]
```



43,159



4,644

Follow @bestiejs

Tweet

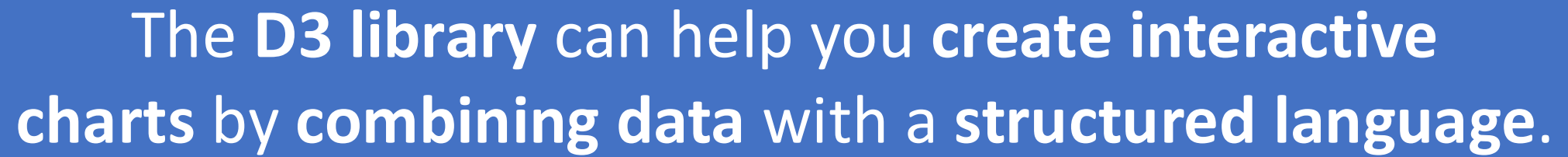
Lodash, makes JavaScript easier by taking the hassle out of working with arrays, numbers, objects, strings, etc.

Lodash is released under the [MIT license](#) & supports modern environments.
Review the [build differences](#) & pick one that's right for you.

Installation

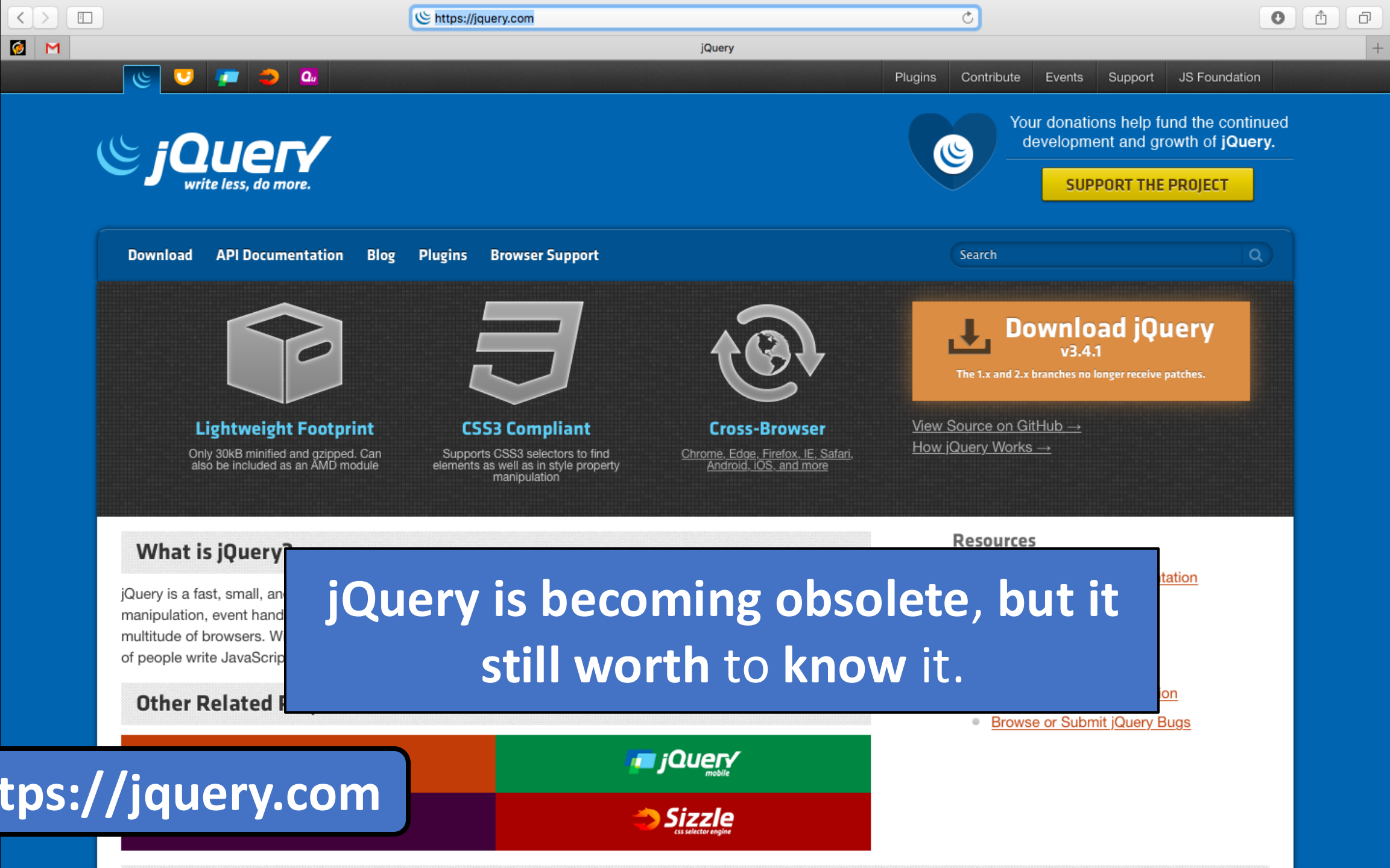
In a browser

<https://lodash.com>



<https://d3js.org>

[See more examples.](#)



Your donations help fund the continued development and growth of jQuery.

SUPPORT THE PROJECT

[Download](#) [API Documentation](#) [Blog](#) [Plugins](#) [Browser Support](#)

Search



Lightweight Footprint

Only 30kB minified and gzipped. Can also be included as an AMD module



CSS3 Compliant

Supports CSS3 selectors to find elements as well as in style property manipulation



Cross-Browser

Chrome, Edge, Firefox, IE, Safari, Android, iOS, and more



Download jQuery

v3.4.1

The 1.x and 2.x branches no longer receive patches.

[View Source on GitHub](#) →

[How jQuery Works](#) →

What is jQuery?

jQuery is a fast, small, and powerful JavaScript library for DOM manipulation, event handling, and Ajax. It works on a multitude of browsers. With jQuery, you can write less code and do more things.

Other Related Resources

Resources

• [Browse or Submit jQuery Bugs](#)

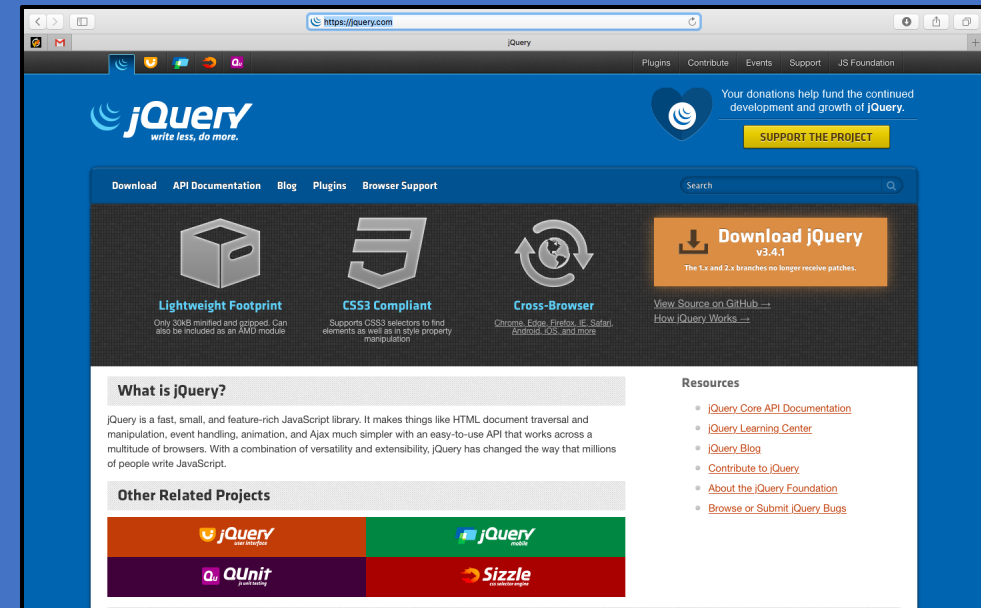
jQuery is becoming obsolete, but it still worth to know it.

<https://jquery.com>



jQuery

- JQuery is one of the most popular library in use today, but it's starting to show its age.
- JQuery gives you easy access to the DOM.
- JQuery is still something front end developers should be comfortable using.





New in v5.2

CSS variables, responsive offcanvas, new utilities, and more!



<https://getbootstrap.com>

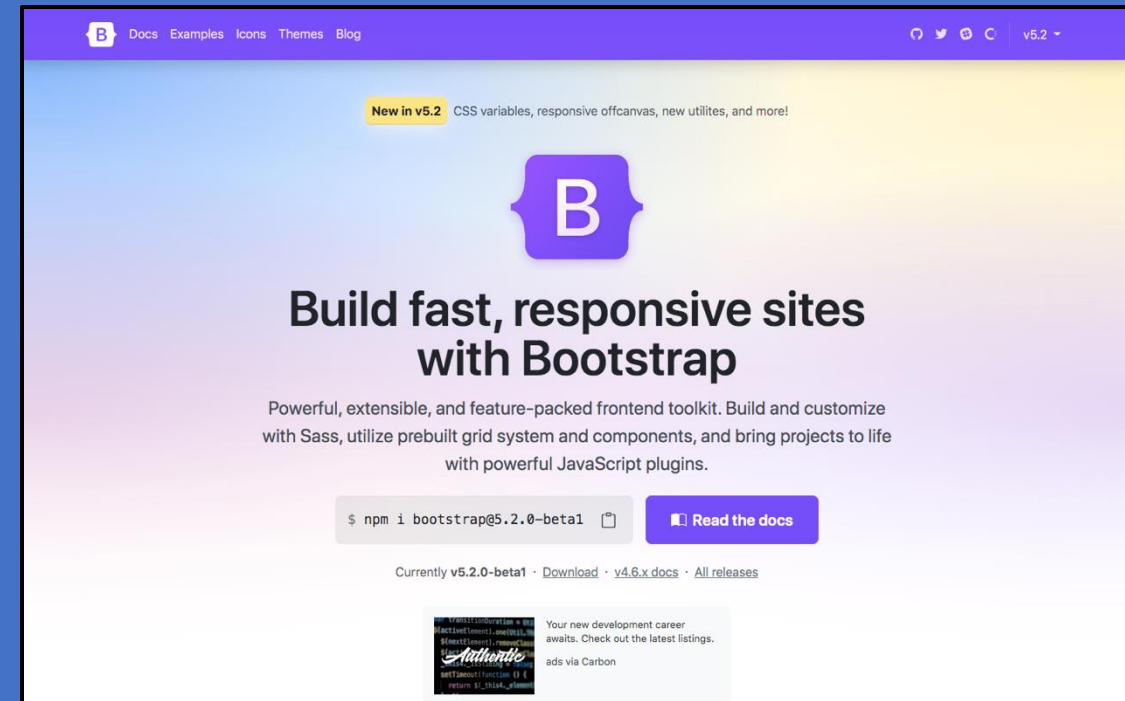
Build fast, responsive sites with Bootstrap

Powerful, extensible, and feature-packed frontend toolkit. Build and customize with Sass, utilize prebuilt grid system and components, and bring projects to life with powerful JavaScript plugins.

Bootstrap is a way to write HTML that adds popular components, like carousels, modals, and cards, making easier to build websites.

Bootstrap framework

- By adding just a few classes in your HTML, you get components with pre-written JavaScript to handle common interactivity.
- There are plenty of similar frameworks, like Material Design and Foundation, but Bootstrap is both the oldest and most popular.



Angular, React.js and Vue.js

Angular, React.js and Vue.js

- Frameworks that help you write in a language that translates to JavaScript.
- A lot of companies will hire developers who specialize in one of these frameworks.



The modern web developer's platform

[GET STARTED](#)

Angular is so large that
is considered a **platform**.

DEVELOP ACROSS ALL PLATFORMS



...one way to build applications with Angular and
reuse your code and abilities to build apps for any
deployment target. For web, mobile web, native mobile
and native desktop.

<https://angular.io>

SPEED & PERFORMANCE

React

A JavaScript library for building user interfaces

[Get Started](#)

[Take the Tutorial >](#)

Declarative views make your code more predictable and easier to debug. **React.js is currently the most popular library.**

React makes it painless to create interactive UIs. Design simple views for each state in your application, and React will efficiently update and render just the right components when your data changes.

Declarative views make your code more predictable and easier to debug.

Build encapsulated components that manage their own state, then compose them to make complex UIs.

Since component logic is written in JavaScript instead of templates, you can easily pass rich data through your app and keep state out of the DOM.

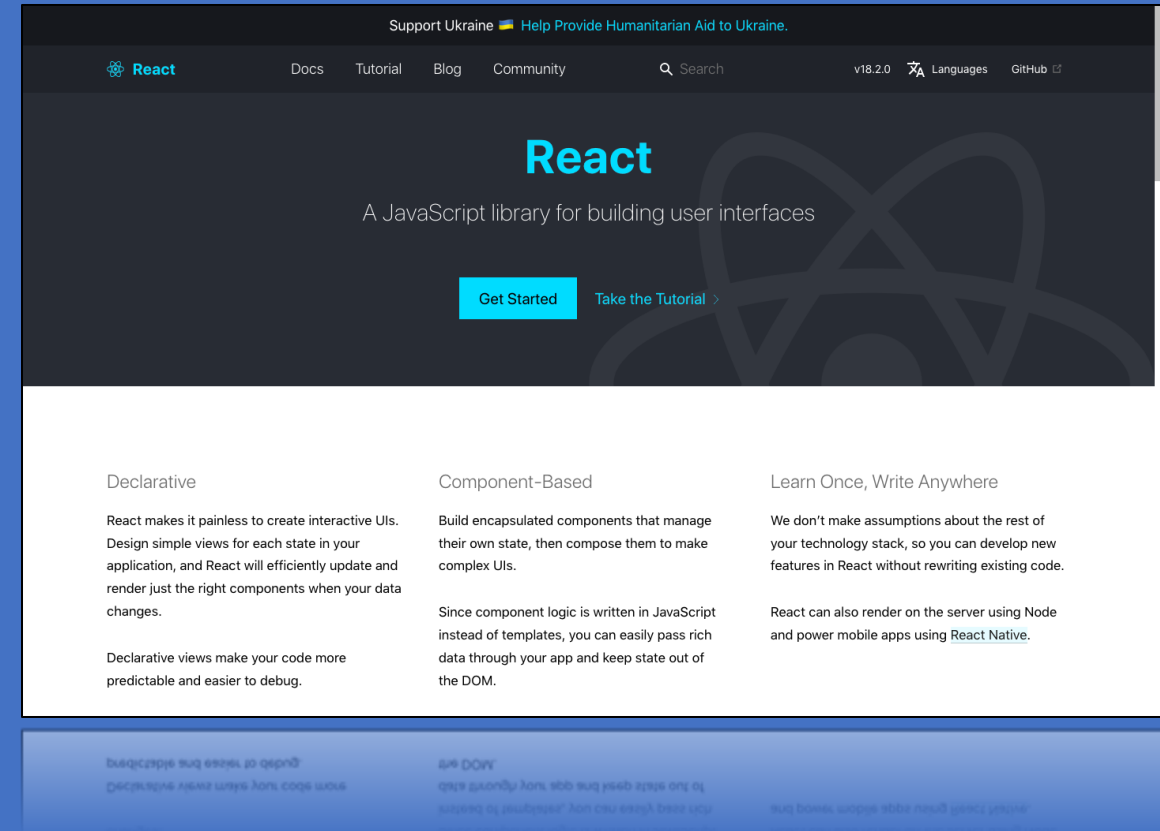
We don't make assumptions about the rest of your technology stack, so you can develop new features in React without rewriting existing code.

React and

<https://reactjs.org>

React.js

- React was designed to create interfaces, so it's less comprehensive than Angular.
- React brings important concepts like a component-based architecture and the virtual DOM, which manages changes in the pages for you.



The **Progressive** JavaScript Framework

An approachable, performant and versatile framework for building web user interfaces.

**Vue.js is the new kid on the block.
Similar to React, although easier to learn and implement.**

Special Sponsor  **HBuilder** Advanced IDE for Vue

Approachable

Builds on top of standard HTML, CSS and JavaScript with intuitive API and world-class documentation.

Performant

Truly reactive, compiler-optimized rendering system that rarely requires manual optimization.

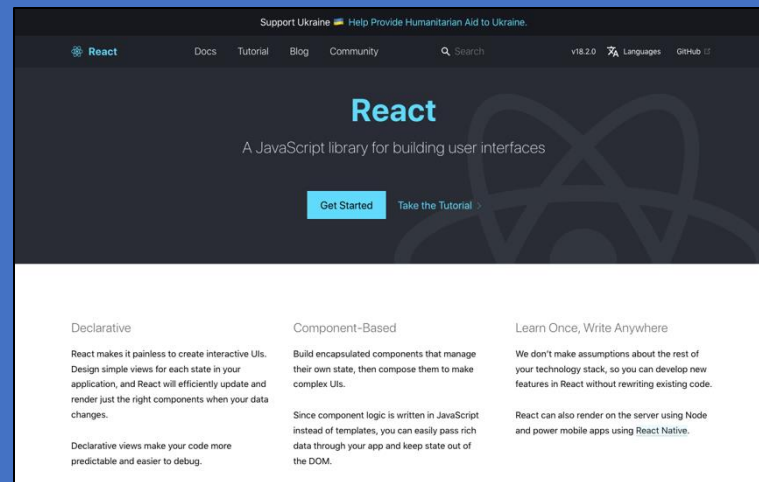
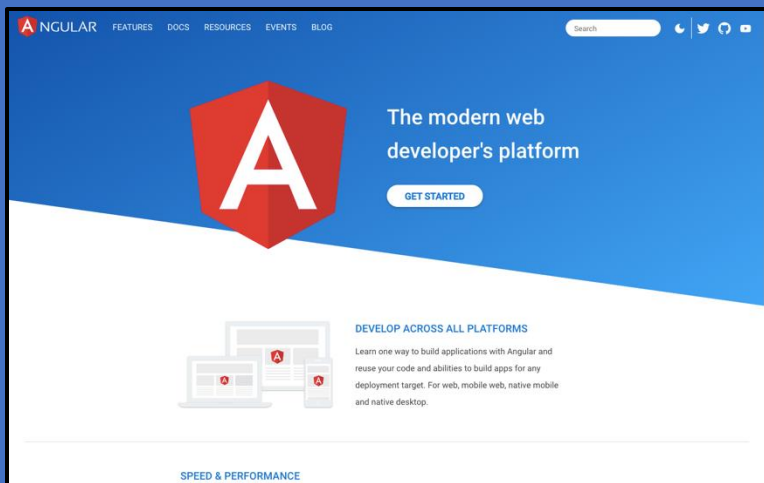
Versatile

A rich, incremental, and adoptable ecosystem that scales between small and large full-featured frameworks.

<https://vuejs.org>

Angular, React.js and Vue.js

- To be a **marketable developer**, you should be **comfortable** with at **least one** of these **frameworks**.
- If you **know one** of these **frameworks**, you can easily **move** your **skills** to the **others**.



Transpiled languages

Lecture: Transpiled languages

Module: Front-end development tools, frameworks & libraries

Transpiler

- **Developers** also have to **worry** about **browser support issues** because the **language** has **matured rapidly** and **browsers don't** always **keep up** with the **latest improvements**.
- To **take care** of this, most **developers use** something called a **transpiler**.
 - ✓ A **transpiler** will **convert code written** in newer **versions of JavaScript** so that they work in **older versions**.

Transpiled languages

- A **Sass code** generates an infinite amount of **CSS code** for **buttons based on a list of colors**.
 - ✓ You could spend more time writing and updating your **HTML, CSS, and JavaScript**, or you could **use a language** like **Sass** that **gets compiled** into **CSS**.
- A lot of languages, like **Sass**, are ways of **simplifying one** of the **core languages**.
 - ✓ These **languages** will then **translate** into **HTML, CSS, or JavaScript**.
 - ✓ Sometimes, we call these **transpiled languages**.

Black Lives Matter

Sass stands with the protesters against police violence. We encourage our users to **get in the streets and join them if you can.**

[Install](#)[Learn Sass](#)[Blog](#)[Documentation](#)[Get Involved](#)  

CSS with superpowers

Sass is the most mature, stable, and powerful professional grade CSS extension language in the world.



Current Releases: [Dart Sass 1.53.0](#) [LibSass 3.6.5](#) [Ruby Sass](#)  [Implementation Guide](#)

<https://sass-lang.com/>

<https://sass-lang.com/> CSS Compatible

Feature Rich

Mature



It's CSS, with just a little more.

Use with Node.js:

```
npm install -g less  
> lessc styles.less styles.css
```

Or the browser:

```
<link rel="stylesheet/less" type="text/css" href="styles.less" />  
<script src="https://cdn.jsdelivr.net/npm/less@4" ></script>
```

<https://lesscss.org>

Babel is a JavaScript compiler.

Use next generation JavaScript, today.

Babel 7.18 is released! Please read our [blog post](#) for highlights and [changelog](#) for more details!

Put in next-gen JavaScript

```
element.index ?? -1;
```

Get browser-compatible JavaScript out

```
var _element$index;  
  
(_element$index = element.index) !== null ? _e
```

Learn more about Babel with our [getting started guide](#) or check out some [videos](#) on the concepts behind it.

<https://babeljs.io>

Transpiled languages

- Other **problem** these languages can **solve** is how to be **backwards compatible** with **older browsers**.
- By using **transpiled languages**, you can use the **latest versions** and **ideas available** in **newer languages**, and **use tools that convert that code**, so that it **works in older browsers**.

Transpile and build process

Transpile and build process

- When working with **web projects**, you often code in a **version** of a **language** that **needs** to be **transpiled**.
- To do this **developer** use **build processes** to **automate** that **conversion**.

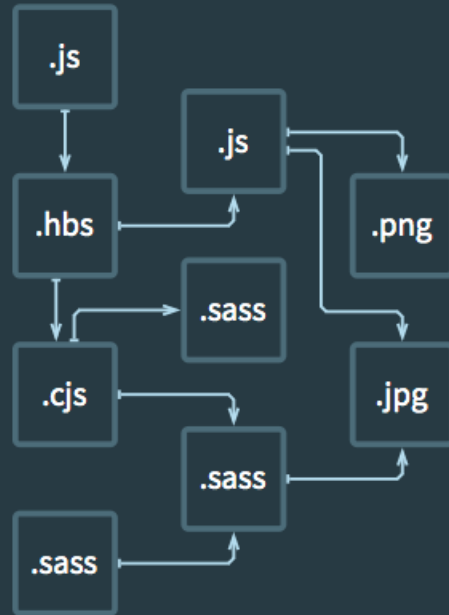
Transpile and build process

- A **build process** is a **set of automation commands** that take care of **common tasks**.
 - ✓ They include **tools** like **Webpack**, which can **bundle** your **code** and **remove unused pieces**.
 - ✓ **Babel**, which **transpose** your **code** for **older browsers** and **local servers** so you can **test the application** on your **own machine**.

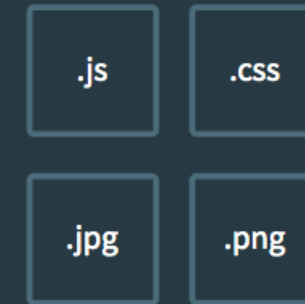
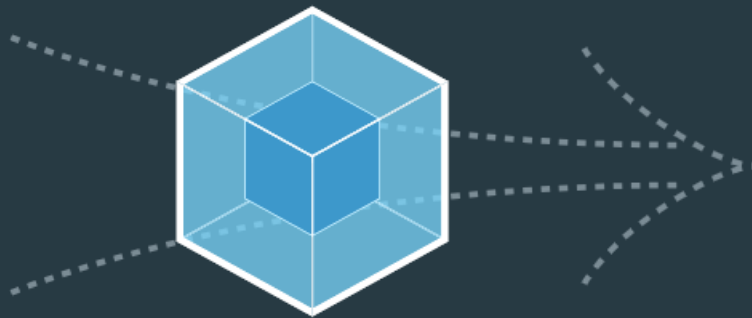
Transpile and build process

- **Build processes** make **development easier** and can also be a **powerful tool** for **automating** and **testing** your **code**.
- **Learning** to take advantage of these **developer tools** is an important **piece** and helping you become a **more productive developer**.

bundle your scripts



MODULES WITH DEPENDENCIES



STATIC ASSETS

<https://webpack.js.org>

Write Your Code

JavaScript

- **JavaScript** has also been **implemented** on the **server side** through something called **Node.js**.
 - ✓ An **implementation** of the **engine** that **runs** in most **browsers** called **Chrome V8**.
 - ✓ This allows you to use **JavaScript** in **unexpected ways**.
 - For **example**, it gives you access to the **file system**.



Source: <https://bit.ly/3nqz919>

Can I use

? ⚙ Settings

Index of features

▼ Filter features

Latest features

- [CSS Container Query Units](#)
- [Declarative Shadow DOM](#)
- [CSS font-palette](#)
- [COLR/CPAL\(v1\) Font Formats](#)
- [CSS @when / @else conditional rules](#)

Most searched features

1. [CSS Grid](#)
2. [Flexbox](#)
3. [WebP image format](#)
4. [gap property for Flexbox](#)
5. [ES6](#)

Test a feature

Our partnership with [BrowserStack](#) now lets you test your website for compatibility across 2,000+ real browsers and devices.

Test on:

- [IE 11](#)
- [Safari on iPhone 11 Pro](#)
- [Safari 13](#)
- [Chrome on Galaxy S20](#)

Did you know?

If a feature you're looking for is not available on the site, you can [vote to have it included](#). Better yet, if you've done the research you can even [submit it yourself](#)!

Third party tools

- [The CanIUse Embed](#) — Add support tables to your site
- [Caniuse Component](#) — Add support tables to your presentations
- [Caniuse command line tool](#)
- [Doiuse...?](#) — Lint your CSS to check what features work
- [I want to use](#) — Select multiple features and see what % of users can use them

[See full list](#)

Browser scores

[Current version](#) [Dev version](#)[Chrome 103: 400](#)[Firefox 102: 379](#)[Safari 15.5: 356](#)<https://caniuse.com>

Can I use

Fetch

? ⚙ Settings

50 results found

☒ Caniuse (3)☒ MDN (47)

Fetch 📄 - LS

Usage

% of all users

Global

94.66% + 0.04% = 94.7%



A modern replacement for XMLHttpRequest.

Current aligned

Usage relative

Date relative

Filtered

All



IE	Edge	Firefox	Chrome	Safari	Opera	Safari on iOS	Opera Mini	Android Browser	Opera Mobile	Chrome for Android	Firefox for Android	UC Browser for Android	Samsung Internet	QQ Browser	Baidu Browser	Kait Brow
		2-33	4-39		10-26											
		1 4 34-38	2 40		2 27											
	12-13	4 39	2 3 41	3 1-10	2 3 28	3 2-10 2										
6-10	14-102	40-101	42-102	1				2.1-4.4.4	12-12.1				4-16.0			
11	103	102	103					103	64	103	101	12.12	17.0	10.4	7.12	2..
		103-104	104-106													

Chrome 104 - 106

Support info

Browser versions

✓ Supported

Released

Total usage

Global: 0.01%

[Test on Chrome 104 - 106](#)

Notes

Test on a real browser

Su

Resources (7)

Feedback

¹ Partial support can be enabled in Firefox with the `dom.fetch.enabled` flag.

² Partial support can be enabled in Chrome with the `dom.fetch.enabled` flag.

Can I use

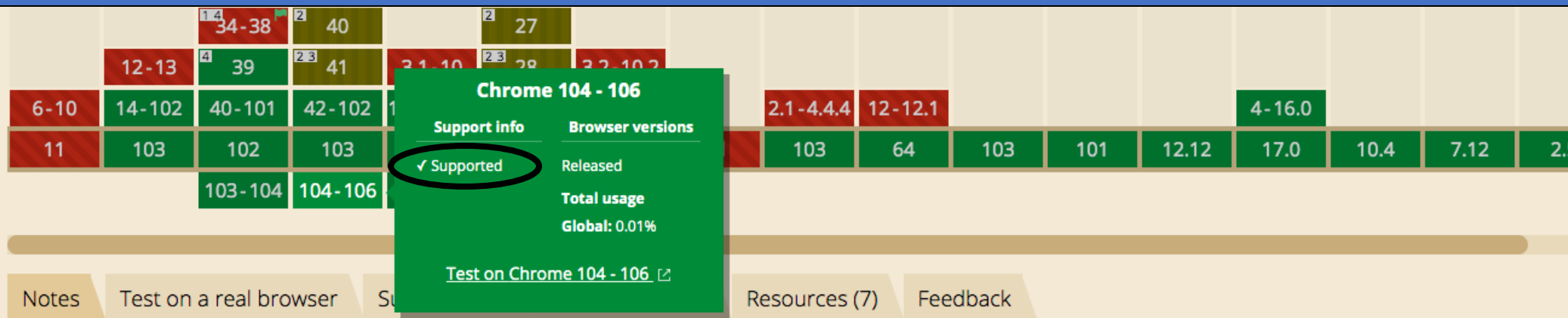
Fetch

? ⚙ Settings

50 results found

☒ Caniuse (3)☒ MDN (47)

Fetch is a relative new JavaScript feature that eases the development of asynchronous requests with AJAX, eliminating the need to handle XMLHttpRequest “by hands”. As we can see, it is well supported nowadays.



¹ Partial support can be enabled in Firefox with the `dom.fetch.enabled` flag.

Questions?